

MODULE 3: PROBABILITY AND DISTRIBUTIONS

1. Probability Foundations

1.1 Random Experiment

A random experiment is one whose outcome cannot be predicted with certainty.

Example: Tossing two coins.

1.2 Sample Space (S)

The sample space is the set of all possible outcomes.

$S = \{HH, HT, TH, TT\}$

$P(S) = 1$

1.3 Event

An event is any subset of the sample space.

$A = \{HH\}$, $P(A) = 1/4$

$B = \{HH, HT, TH\}$, $P(B) = 3/4$

2. Probability Rules

Conditional Probability:

$P(A|B) = P(A \cap B) / P(B)$

Bayes Rule: $P(A|B) = [P(B|A)P(A)] / P(B)$

Independence: $P(A \cap B) = P(A)P(B)$

3. Random Variables

$X : S \rightarrow R$

Discrete RV: $P(X=x)=f(x)$, where $\sum f(x)=1$

Continuous RV: $f(x) \geq 0$ and $\int f(x)dx=1$

Mean (Discrete): $E(X)=\sum x f(x)$

Variance: $Var(X)=E(X^2) - [E(X)]^2$

4. Binomial Distribution

$P(X=x) = {}^nC_x p^x (1-p)^{(n-x)}$

Mean = np
Variance = $np(1-p)$

5. Poisson Distribution

$P(X=x) = e^{-\lambda} \lambda^x / x!$
Mean = λ
Variance = λ

6. Geometric Distribution

$P(X=x) = (1-p)^{x-1} p$
Mean = $1/p$
Variance = $(1-p)/p^2$

7. Normal Distribution

$f(x) = 1/\sqrt{2\pi\sigma^2} e^{-\{(x-\mu)^2/(2\sigma^2)\}}$
Mean = μ
Variance = σ^2